1269–79

1280–99

“YOU WILL BE ABLE TO **DIRECT YOUR STEPS** TO CITIES AND ISLANDS AND TO **ANY PLACE** WHATEVER IN THE **WORLD.**”

Pierre de Maricourt, French scholar, describing the compass in *Epistola de Magnete*, 1269

Figures wearing eyeglasses soon found their way into religious art, as is shown here in this 1491 detail from the *Betrayal by Judas* in Notre Dame, Paris, France.

Gaocheng observatory
One of its two towers contained an armillary sphere. Between them lay a "sky-measuring scale" to measure the shadow of the 39-foot gnomon.

IN 1269, FRENCH SCHOLAR PIERRE DE MARICOURT wrote *Epistola de magnete* (*Letter on the Magnet*), the first work to describe the **properties of magnets**. In it, he set out the laws of magnetic attraction and repulsion, and explained how to identify the poles of compasses. Maricourt's work led to the construction of **better magnetic compasses**, which became invaluable aids in sea navigation. He also described the operation of a **perpetual motion machine**, which worked using magnetism.

In 1276, the Mongol ruler of China, Kublai Khan, asked mathematician and engineer **Guo Shoujing** (1231–1316) to reform the calendar. To perform this task, Guo first had a series of astronomical instruments constructed. This included a vast equatorial armillary sphere — calibrated with a ring representing equatorial coordinates, a system not used in Europe until the time of Tycho Brahe (see 1565–74) three centuries later. He then established **astronomical observatories** at Peking (Beijing) and Gaocheng, near Loyang, China, between 1279 and 1280. At the latter, a 39 ft- (12 m-) high gnomon (shaft on a sundial) sat on top of a pyramid, casting shadows, which were measured at the time of the Sun's solstices to help determine the **length of the year**. Guo used advanced trigonometry to calculate the length of a year.

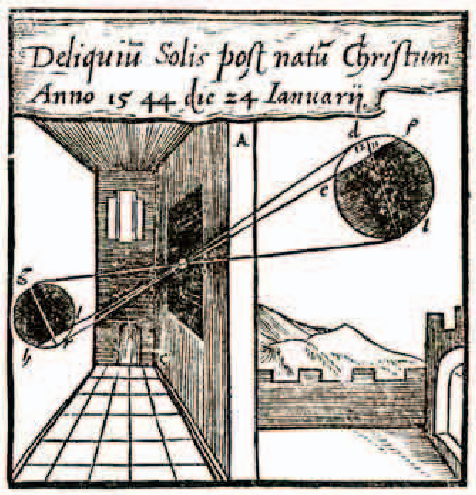
IN 1280, GUO SHOUJING finally completed his **calendar**. According to his calculations, a year had 365.2425 days.

The **earliest surviving cannon** is from China and was made *c.*1288. Before cannons were strengthened by the use of cast iron barrels, the Chinese had probably used bronze tubes to eject projectiles using gunpowder explosives. Manuscripts of 1274 and 1277, however, refer to *huo pa'o*—explosive weapons used by the Mongols to demolish the ramparts of Chinese cities—so the invention may have occurred a little earlier.

Although the magnifying properties of glass lenses had been studied by English bishop Robert Grosseteste (1175–1253) and Roger Bacon earlier in the 13th century, the **first description of eyeglasses** was given by a Dominican friar **Giordano da Pisa** (*c.*1260–1310), who wrote that he had seen them in 1286. Early glasses were convex (curving out) to correct far-sightedness. Concave lenses (curving in) for near-sightedness did not appear for more than a century.

In 1290, French astronomer **William of St. Cloud** gave an account of a **solar eclipse** witnessed by him five years earlier. Many of those who had observed the eclipse had damaged their eyes by viewing the Sun directly. In order to prevent this, William used a **camera obscura**—a type of pinhole camera in which light goes into a dark chamber and is projected through a tiny aperture onto another surface, such as a card, opposite it. The technique had been used by earlier astronomers, such as Alhazen in the 11th century, to prove that intersecting rays do not interfere with each other. William, however, was the **first to explain its use in solar observation**. He also calculated an accurate value by which the Earth tilts on its axis, by observing the Sun's position at the solstices. In addition, he produced an almanac with detailed positions of the Sun, Moon, and planets at various dates between 1292 and 1312.

26 SECONDS THE **DIFFERENCE** BETWEEN **GOU'S CALCULATION** AND **ACTUAL DAYS IN A YEAR**



Camera obscura
This 16th-century illustration of a camera obscura shows how an image of the Sun is reversed after light passes through an aperture onto a surface in a darkened room.

1269 William of Moerberke translates works of **Archimedes** into Latin

1269 Pierre de Maricourt describes **properties of magnets**

1270–78 Polish monk **Witelo** publishes *Perspectiva*, on optics

1274 First account of use of **cannon** in China

1275 William of Saliceto recommends use of **surgical knife** rather than cauterization

c.1275 Bernard of Verdun defends **Ptolemy's theory** of epicycles

1276 Guo Shoujing begins work to reform the **Chinese calendar**

1279–80 Guo Shoujing constructs **observatories** at Peking and Gaocheng

1286 Giordano da Pisa gives earliest description of **eyeglasses**

c.1288 Earliest surviving **hand-cannon** from China

1290 William of St. Cloud describes the use of **camera obscura** to observe the Sun indirectly during an eclipse

1296 Use of Alfonsine Tables—astronomical tables—spreads to **Paris**